APPLIED MACHINE LEARNING AND ECONOMETRICS CASE

Korea Income & Welfare

This project studies how education relates to income in Korean welfare microdata. The updated version keeps the strongest part of the original delivery, which was its theoretical motivation, and expands it with formal tests, centered-age econometrics, distributional analysis, and cross-validated machine-learning benchmarks.

89,935
Valid records in the cleaned analytical layer

86,313
Trimmed modeling sample used for prediction and econometrics

0.645
Best holdout R-squared from the tuned random forest

12.47%
Average income premium per education step in OLS

Research motivation

The original notebook framed education and income as a serious social question rather than as a generic predictive exercise. That framing remains essential. Education is not only a variable that helps fit a model. It stands in for human-capital formation, credentials, labor-market sorting, and differential access to opportunity.

The upgraded version therefore asks a richer sequence of questions. First, does education visibly separate income outcomes in the raw data? Second, how much predictive power is gained when broader demographic and household context is included? Third, is the return to education constant across the income distribution or does it rise in better-paid parts of the sample?

> Executive takeaway: the project is not just predicting income. It is estimating whether education has a stable economic premium and whether that premium changes across the distribution.

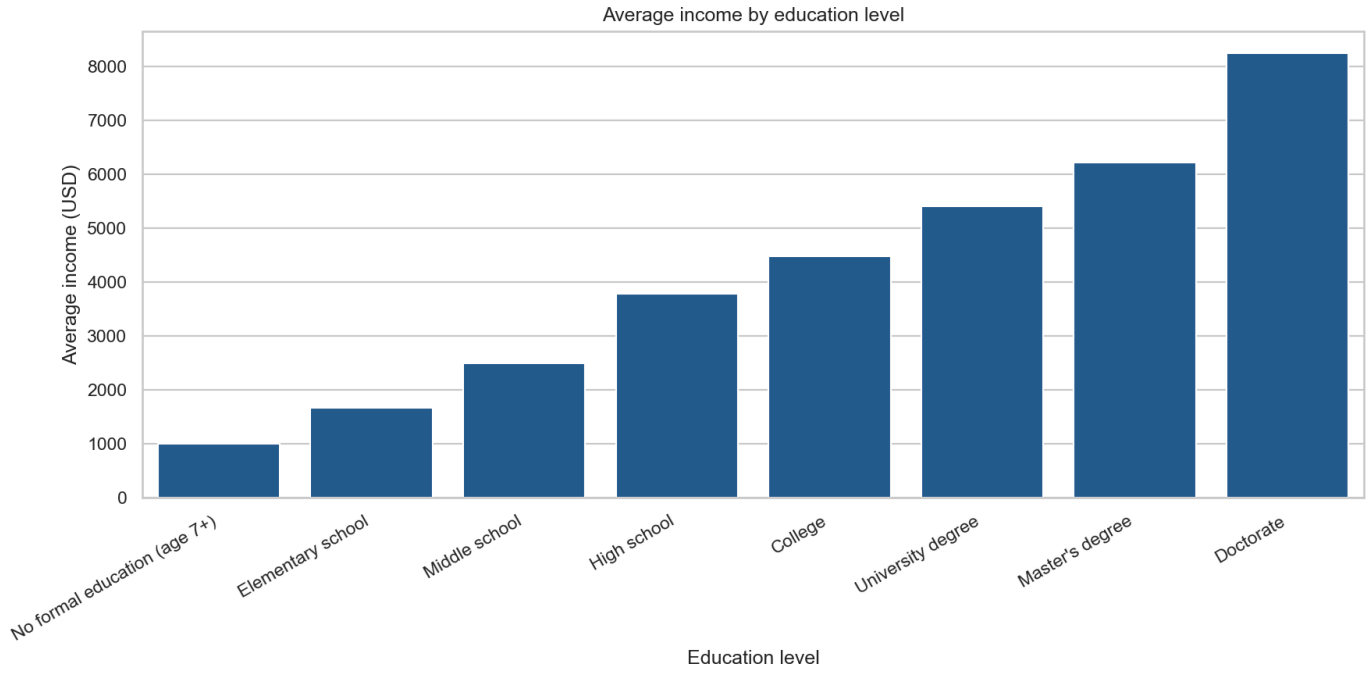
Data architecture and descriptive base

The dataset begins with `92,857` observations and is cleaned into a valid analytical sample of `89,935` records. For the predictive and econometric layers, extreme outliers are trimmed to produce a stable working sample of `86,313` records. This is large enough to support both flexible ML models and coefficient-based inference.

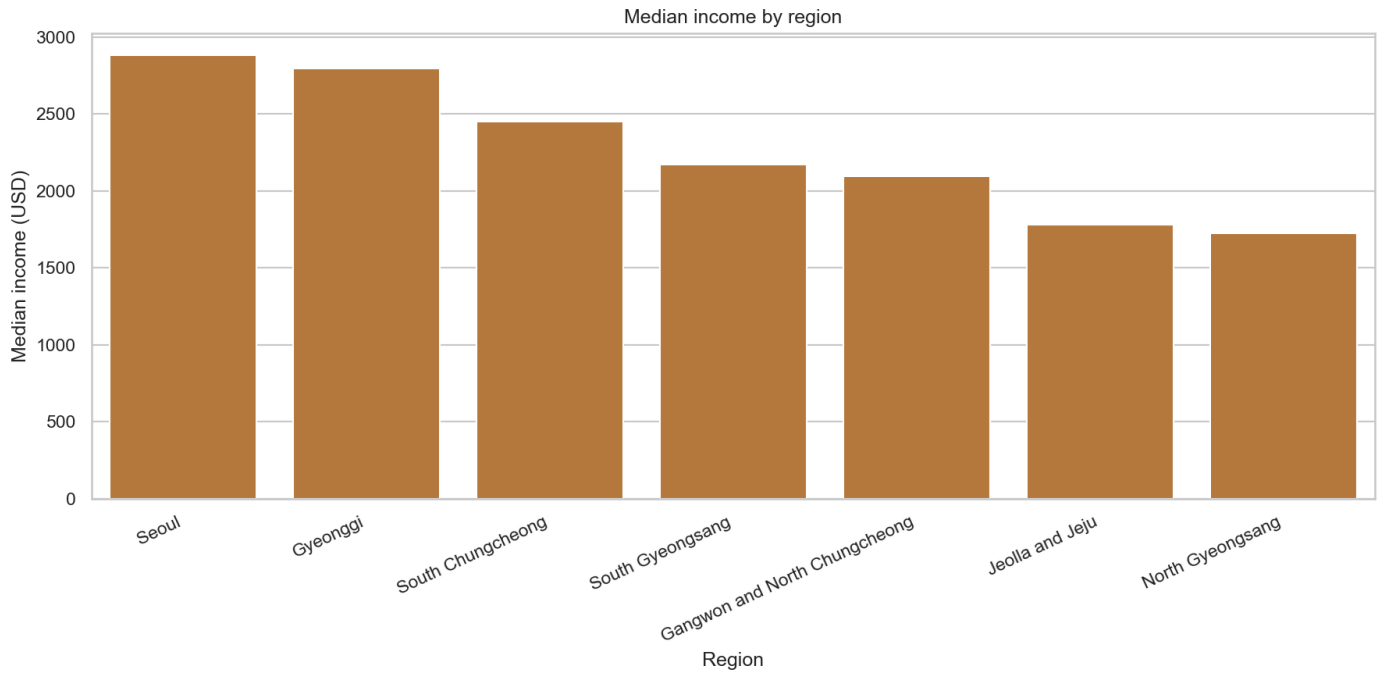
The descriptive evidence already tells a coherent story. Income rises steadily across education levels, varies across regions, and changes over time. These patterns do not prove a mechanism on their own, but they do justify moving into more formal models.

Baseline signals

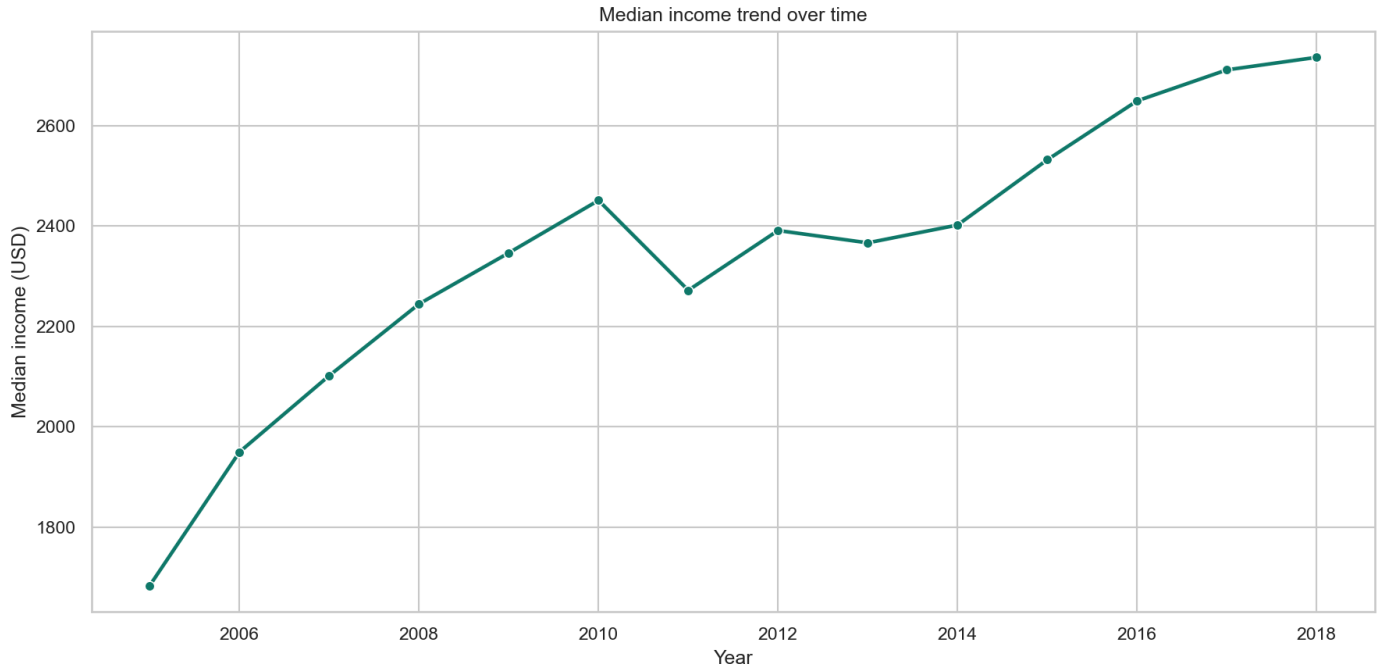
- Income differences across education levels are very large.
- Seoul is the top region by median income in the cleaned sample.
- Household size remains a major part of the predictive story.
- Gender differences in raw income remain visible before controls.



Income increases across the education ladder before any regression is estimated. This makes the education-income question both intuitive and empirically defensible.



Regional structure matters too. Education is central, but it is not the only axis along which income differs.



The time dimension matters because part of the income story reflects changing macro context, not only individual characteristics.

Formal testing of the descriptive story

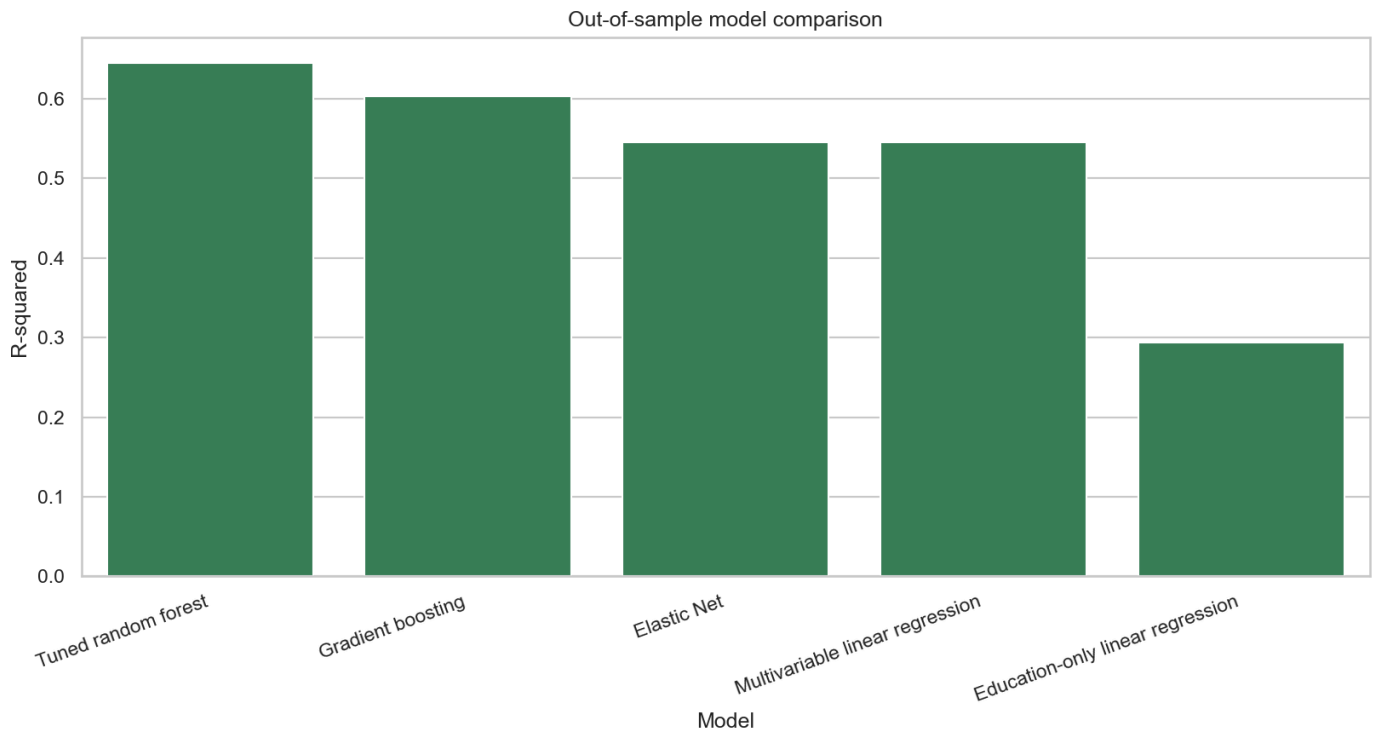
The upgraded project no longer treats the descriptive layer as self-evident. It now tests whether the key contrasts are statistically credible.

Question	Test	Result	Interpretation
Do income distributions differ across education levels?	Kruskal-Wallis	Statistic `33,934.36`, `p < 0.001`	Education levels map into clearly different income distributions.
Is education monotonically associated with income?	Spearman correlation	`rho = 0.609`, `p < 0.001`	The education-income relationship is not weak or ambiguous; it is strongly positive in rank terms.
Do men and women differ in raw income distributions?	Mann-Whitney U	`p < 0.001`	Gender differences remain visible before adding controls and functional form.

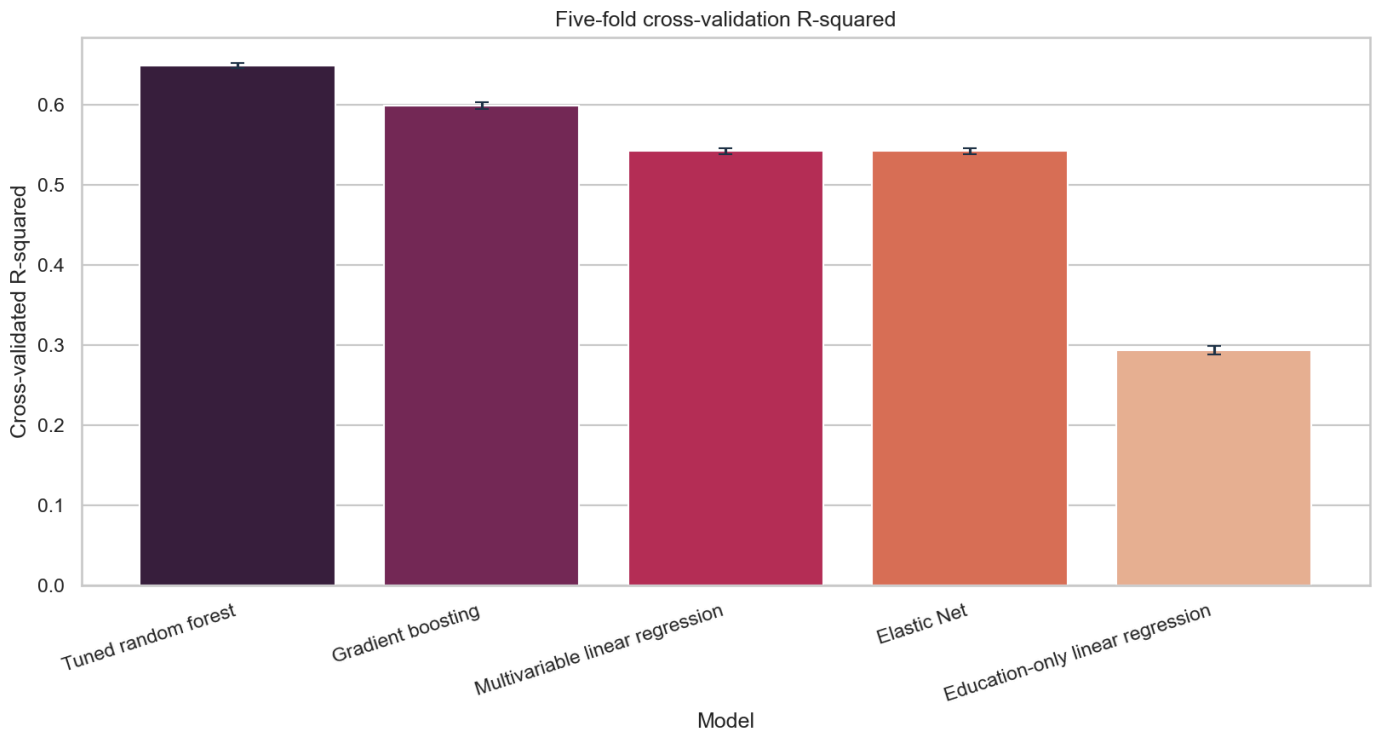
> This matters because the project now shows a proper sequence: descriptive pattern, formal test, predictive benchmark, then econometric interpretation.

Predictive benchmarking

The machine-learning layer answers a practical question: how much predictive performance is gained when we move from a very simple education-only model to richer multivariable and nonlinear specifications?

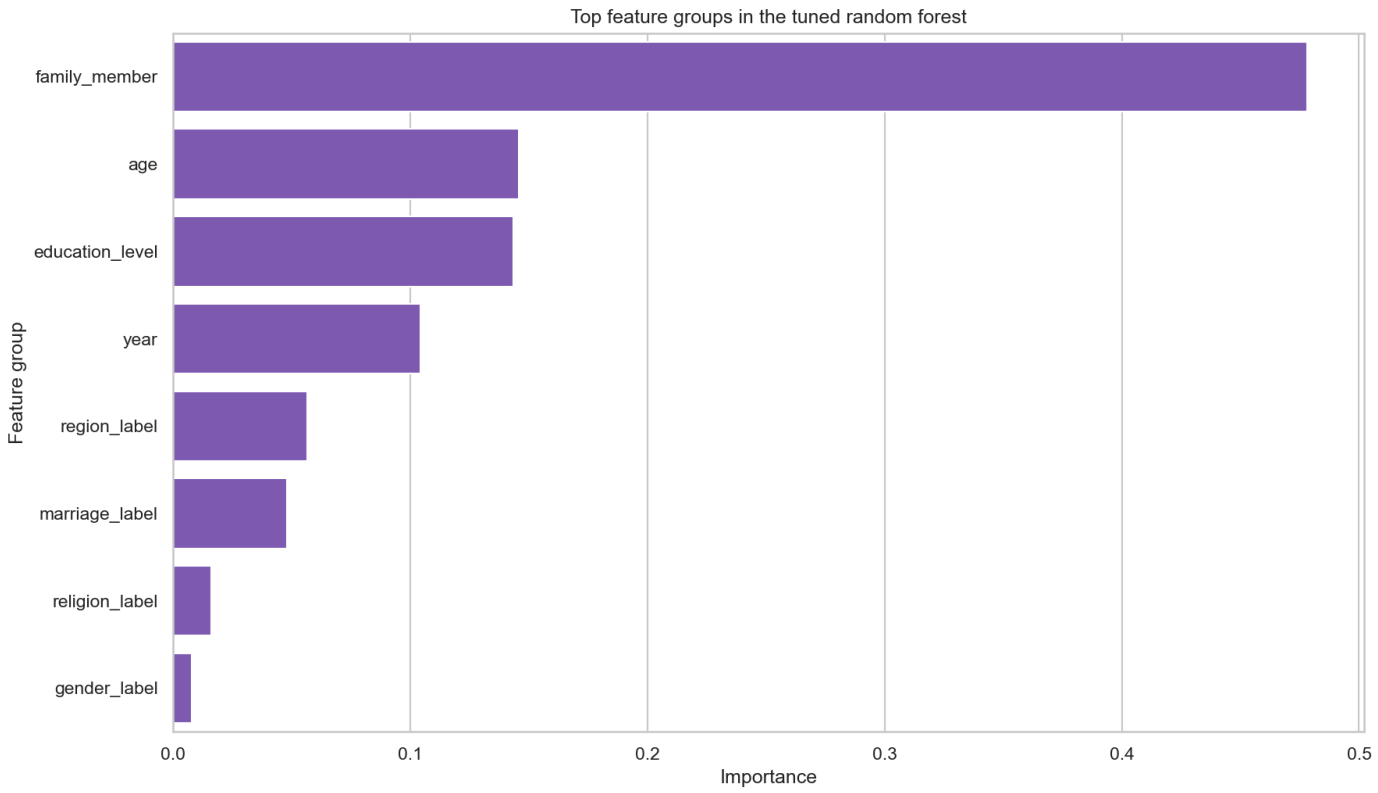


The tuned random forest is the strongest predictive model, but the comparison only becomes convincing once it is paired with cross-validation and interpretation.

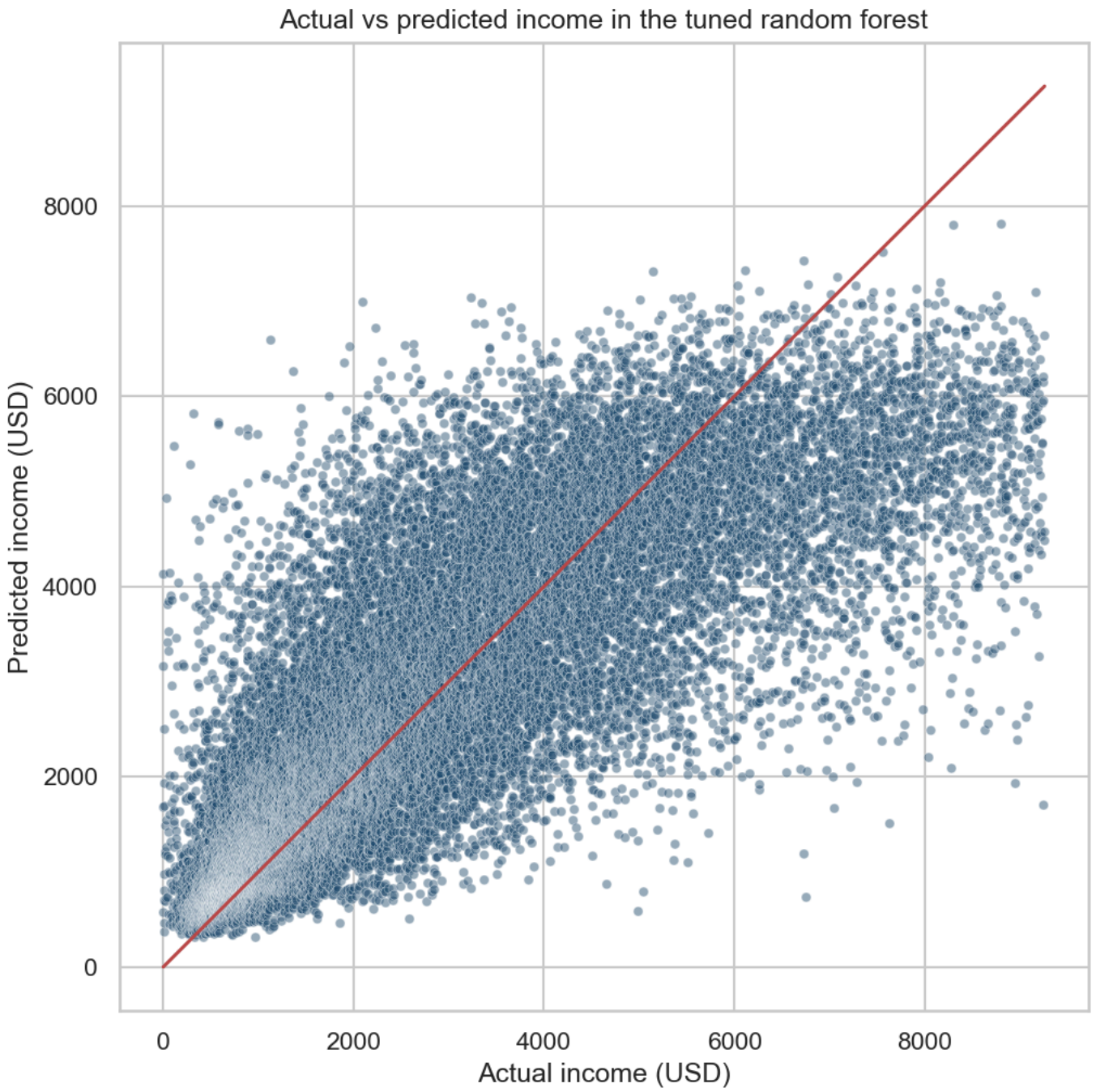


Five-fold cross-validation confirms the holdout ranking. The tuned random forest remains strongest with `CV R^2 = 0.649`, well above the education-only baseline of `0.294`.

Model	Holdout R-squared	Cross-validated R-squared	Interpretation
Tuned Random Forest	`0.645`	`0.649`	Best predictive benchmark and stable across folds.
Gradient Boosting	`0.603`	`0.599`	Strong nonlinear alternative, but still below the tuned forest.
Multivariable Linear Regression	`0.545`	`0.542`	Interpretability comes with a clear but not catastrophic performance tradeoff.
Education-only Linear Regression	`0.294`	`0.294`	A useful benchmark that shows how much context beyond education matters for prediction.



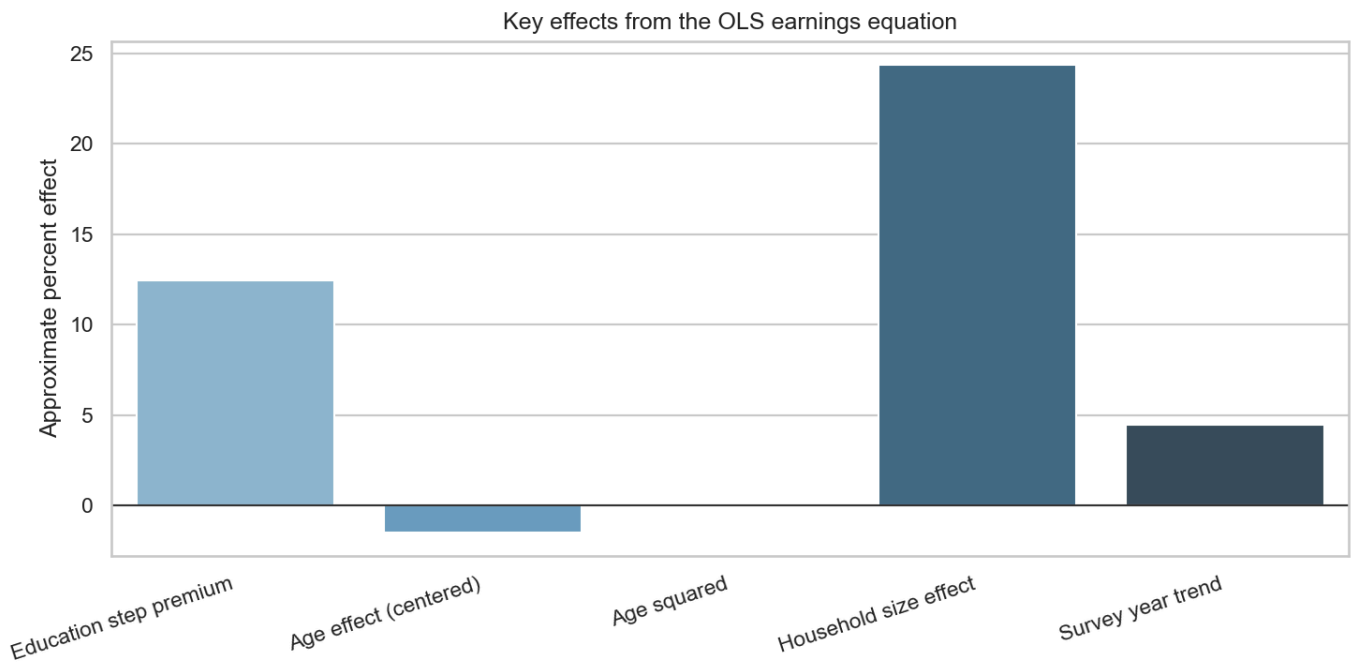
The tuned random forest identifies household size, age, education, and year as the most important feature groups. The predictive layer broadens the story without making education disappear.



The best model tracks the middle of the income distribution well, while still facing the usual challenges at the upper tail. This is a realistic predictive pattern rather than an overfit success story.

Econometric interpretation

The OLS layer is where the project becomes easier to interpret in professional settings. The semilog specification estimates the average income premium per education step while controlling for age, household size, region, gender, marital status, religion, and year. Age is centered before adding the quadratic term, which sharply improves collinearity metrics.

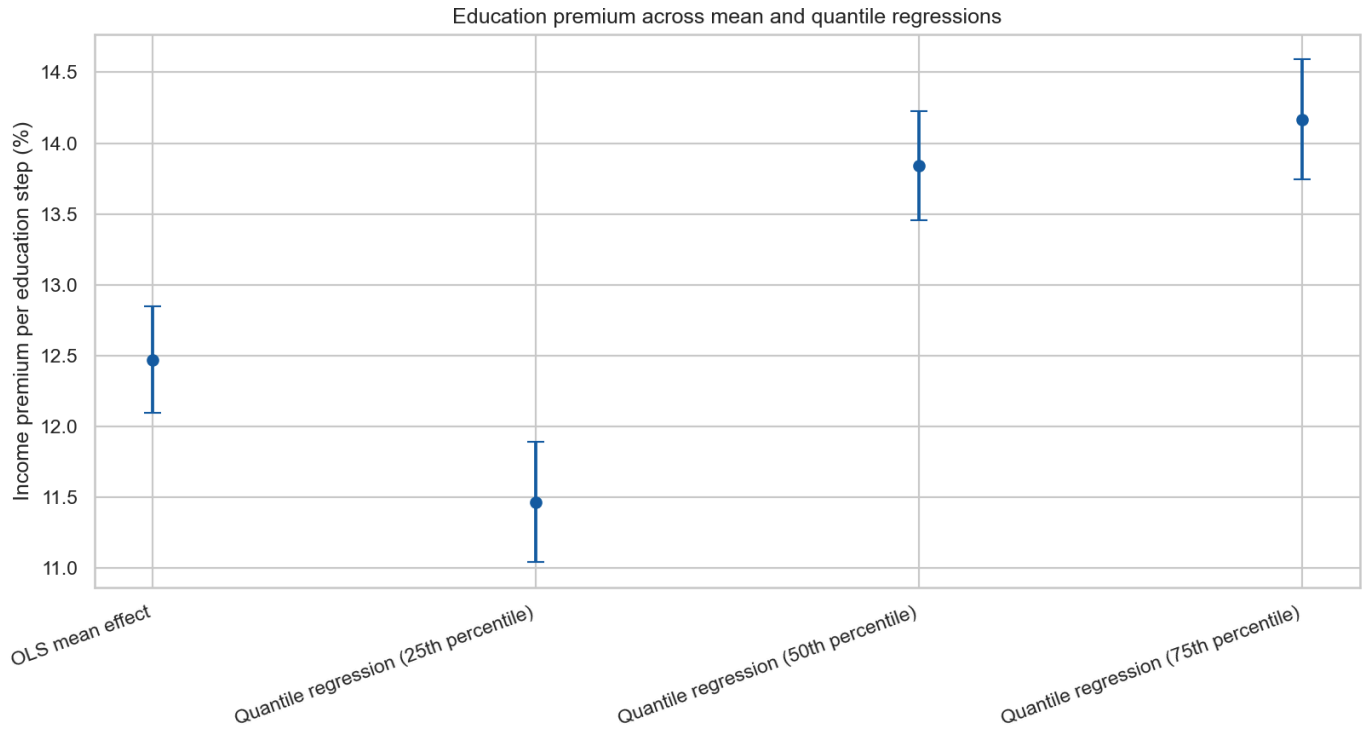


The most policy-relevant average effect is still the education premium. Household size and time trend also matter materially once the full specification is estimated.

Coefficient	Value	Meaning
Education step premium	`12.47%`, `p < 0.001`	Each education step is associated with a sizable increase in income on average.
Household size effect	`24.39%`, `p < 0.001`	Larger households are strongly associated with higher total income, which is why a separate welfare interpretation is still needed.
Adjusted R-squared	`0.600`	The semilog specification explains a large share of variation for a social microdata application.
VIF range	`1.03` to `1.96`	Centering the age polynomial keeps multicollinearity under control.
Breusch-Pagan and RESET	Both strongly significant	The data justify robust standard errors and support keeping a richer nonlinear benchmark next to OLS.

Quantile regression and heterogeneous returns

The most interesting analytical extension is quantile regression. Instead of assuming that schooling pays off equally everywhere in the distribution, the project tests whether the education premium changes across lower-income, middle-income, and upper-income observations.



The education premium rises from `11.46%` near the lower quartile to `14.17%` near the upper quartile. That moves the project from average effect estimation into distributional interpretation.

Model	Education premium	Interpretation
OLS average effect	`12.47%`	Average return across the modeled sample.
25th percentile	`11.46%`	Education matters even in the lower-income part of the sample, but the payoff is smaller.
50th percentile	`13.84%`	The middle of the distribution sees a stronger premium than the lower tail.
75th percentile	`14.17%`	The premium is strongest in better-paid parts of the distribution.

This is the point where the project stops being a generic ML exercise. The quantile result suggests that education interacts with labor-market sorting or access to higher-reward segments rather than producing a completely uniform income effect.

Conclusion and theoretical proposal

The repository now supports a richer theoretical statement. Education has a clearly positive income premium in this welfare dataset, but that premium is heterogeneous. It appears stronger at higher-income quantiles, which suggests that schooling is rewarded more strongly where individuals are already positioned closer to better-paying labor-market segments.

For hiring purposes, this is useful because it shows two different kinds of maturity. First, the project can tune and compare predictive models honestly. Second, it knows when predictive performance is not the whole point, and when coefficient interpretation and distributional heterogeneity matter more than the single best score.

> Hiring takeaway: this is no longer only a machine-learning notebook. It is an applied income-research case with theory, diagnostics, benchmarking, and a stronger substantive conclusion about heterogeneous returns to education.